

Markov Processes: Homework 1

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Repository: <https://github.com/lindenkurtz/Markov2026>

Handwritten notes can also be found in this repo.

Problem 1: Joint distributions

- (a) Due to the nature of probability distributions, $\int_0^2 \int_0^1 K(x^2 + y) dx dy = 1$.

$$\int_0^1 \int_0^2 x^2 dy dx + \int_0^1 \int_0^2 y dy dx = \frac{1}{K}, \quad (1)$$

$$\int_0^1 2x^2 dx + \int_0^1 2 dx = \frac{1}{K}, \quad (2)$$

$$\frac{2}{3} + 2 = \frac{1}{K} \quad (3)$$

Therefore $K = \frac{3}{8}$.

- (b) The marginal of X is $f_X(x) = \int_0^2 f(x, y) dy$ for $0 \leq x \leq 1$, and zero otherwise.

$$f_X(x) = \int_0^2 \frac{3}{8}(x^2 + y) dy = \frac{3}{4}x^2 + \frac{3}{4}$$

As a check, $\int_0^1 f_X(x) dx = \int_0^1 \frac{3}{4}x^2 + \frac{3}{4} dx = 1$.

Similarly, the marginal of Y is $f_Y(y) = \int_0^1 f(x, y) dx$ for $0 \leq y \leq 2$, and zero otherwise.

$$f_Y(y) = \int_0^1 \frac{3}{8}(x^2 + y) dx = \frac{1}{8} + \frac{3}{8}y$$

As a check, $\int_0^2 f_Y(y) dy = \int_0^2 \frac{1}{8} + \frac{3}{8}y dy = 1$.

- (c) X and Y are independent if and only if $f(x, y) = f_X(x)f_Y(y)$ for all (x, y) . Comparing the two expressions,

$$f_X(x)f_Y(y) = \left(\frac{3}{4}x^2 + \frac{3}{4}\right)\left(\frac{1}{8} + \frac{3}{8}y\right), \quad (4)$$

$$= \frac{3}{32}x^2 + \frac{9}{32}x^2y + \frac{3}{32} + \frac{9}{32}y, \quad (5)$$

$$\neq f(x, y) = \frac{3}{8}(x^2 + y) \quad (6)$$

So X and Y are not independent.

Problem 2: Moment generating function 1

- (a) Given $X \sim \text{Geometric}(q)$ counting misses, so $\mathbb{P}(X = x) = (1 - q)^x q$. Therefore, the MGF is given by:

$$M_X(s) = \mathbb{E}[e^{Xs}], \quad (7)$$

$$= \sum_{x=0}^{\infty} e^{xs} (1 - q)^x q, \quad (8)$$

$$= q \sum_{x=0}^{\infty} (e^s (1 - q))^x \quad (9)$$

Assuming that $e^s (1 - q) < 1$, or $e^s < \frac{1}{1-q}$,

$$M_X(s) = \frac{q}{1 - e^s (1 - q)}$$

And the MGF is finite for $s < \ln(\frac{1}{1-q})$.

- (b) Differentiating at $s = 0$:

$$M'_X(s) = \frac{q(1 - q)e^s}{(1 - e^s(1 - q))^2}, \quad (10)$$

$$M'_X(0) = \frac{1 - q}{q}, \quad (11)$$

$$M''_X(s) = \frac{(1 - e^s(1 - q))^2(q(1 - q)e^s) + 2(q(1 - q)e^s)(1 - e^s(1 - q))(1 - q)e^s}{(1 - e^s(1 - q))^4}, \quad (12)$$

$$M''_X(0) = \frac{(1 - q)(q + 2(1 - q))}{q^2}, \quad (13)$$

$$\mathbb{E}[X] = M'_X(0) = \frac{1 - q}{q}, \quad (14)$$

$$\text{Var}(X) = M''_X(0) - (M'_X(0))^2 = \frac{1 - q}{q^2} \quad (15)$$

- (c) Fix an integer $k \geq 0$. Writing the conditional MGF of $X - k$ as an expectation and using $\mathbb{E}[g(X)] = \sum_x g(x)\mathbb{P}(X = x)$,

$$M_{X-k|X \geq k}(s) = \mathbb{E}\left[e^{s(X-k)} \mid X \geq k\right], \quad (16)$$

$$= \sum_{n=k}^{\infty} e^{s(n-k)} \mathbb{P}(X = n \mid X \geq k), \quad (17)$$

$$= \sum_{n=k}^{\infty} e^{s(n-k)} \frac{\mathbb{P}(X = n \cap X \geq k)}{\mathbb{P}(X \geq k)} \quad (18)$$

Every n in this summation already satisfies $n \geq k$, so the event $\{X = n \cap X \geq k\}$ is just

$\{X = n\}$, and $\mathbb{P}(X \geq k) = (1 - q)^k$. Therefore

$$M_{X-k|X \geq k}(s) = \sum_{n=k}^{\infty} e^{s(n-k)} \frac{(1-q)^n q}{(1-q)^k}, \quad (19)$$

$$= \frac{q e^{-sk}}{(1-q)^k} \sum_{n=k}^{\infty} (e^s(1-q))^n, \quad (20)$$

$$= \frac{q e^{-sk}}{(1-q)^k} \cdot \frac{(e^s(1-q))^k}{1 - e^s(1-q)}, \quad (21)$$

$$= \frac{q}{1 - e^s(1-q)} = M_X(s) \quad (22)$$

Since the MGF determines the distribution, this means that the number of further misses for a player who has already missed k shots has the same distribution as if they had shot no shots at all.

- (d) Let $T_r = \sum_{i=1}^r X_i$, where the X_i are i.i.d. random variables with the same distribution as X . Since the X_i are independent, the MGF of the sum is the product of the MGFs:

$$M_{T_r}(s) = (M_X(s))^r = \left(\frac{q}{1 - e^s(1-q)} \right)^r \quad (23)$$

This is the MGF of a negative binomial distribution counting the number of misses before the r th make, with success probability q and r successes. Differentiating,

$$M'_{T_r}(s) = r (q(1 - e^s(1-q))^{-1})^{r-1} \cdot \frac{q(1-q)e^s}{(1 - e^s(1-q))^2}, \quad (24)$$

$$M'_{T_r}(0) = r(1)^{r-1} \cdot \frac{1-q}{q}, \quad (25)$$

$$\mathbb{E}[T_r] = \frac{r(1-q)}{q} \quad (26)$$

and for the second moment,

$$M''_{T_r}(0) = r \cdot \frac{(1-q)(q + 2(1-q))}{q^2} + \frac{1-q}{q} \cdot \frac{r(r-1)(1-q)}{q}, \quad (27)$$

$$= \frac{r(1-q)(q + 2(1-q)) + r(r-1)(1-q)^2}{q^2} \quad (28)$$

so that

$$\text{Var}(T_r) = \frac{r(1-q)(q + 2(1-q)) + r(r-1)(1-q)^2}{q^2} - \frac{r^2(1-q)^2}{q^2}, \quad (29)$$

$$= \frac{r(1-q)(-q + 2 + (r-1)(1-q) - r(1-q))}{q^2}, \quad (30)$$

$$= \frac{r(1-q)(-q + 2 + r - rq - 1 + q - r + rq)}{q^2}, \quad (31)$$

$$= \frac{r(1-q)}{q^2} \quad (32)$$

Both the mean and the variance are exactly r times the corresponding quantities for a single X , which is what a sum of r independent copies should give.

Problem 3: Mixtures and conditioning

Let $\theta \sim \text{Uniform}[0, 1]$ and $X \mid \theta \sim \text{Exponential}(\theta)$. The joint density follows from $f_{X|\theta}(x \mid \theta) = f_{X,\theta}(x, \theta)/f_\theta(\theta)$:

$$f_{X,\theta}(x, \theta) = f_{X|\theta}(x \mid \theta) f_\theta(\theta), \quad (33)$$

$$= \theta e^{-\theta x} \cdot 1, \quad 0 < \theta \leq 1, x > 0 \quad (34)$$

So, every $0 < \theta \leq 1$ contributes to any given x .

Marginalizing over θ and integrating by parts with $u = \theta$ and $v' = e^{-\theta x}$ (so $u' = 1$ and $v = -\frac{1}{x}e^{-\theta x}$),

$$f_X(x) = \int_0^1 \theta e^{-\theta x} d\theta, \quad (35)$$

$$= \left[-\frac{\theta}{x} e^{-\theta x} \right]_0^1 + \int_0^1 \frac{1}{x} e^{-\theta x} d\theta, \quad (36)$$

$$= \left(-\frac{\theta}{x} e^{-\theta x} - \frac{1}{x^2} e^{-\theta x} \right) \Big|_0^1, \quad (37)$$

$$= \left(-\frac{1}{x} e^{-x} - \frac{1}{x^2} e^{-x} \right) - \left(0 - \frac{1}{x^2} \right), \quad (38)$$

$$= \frac{1}{x^2} - e^{-x} \left(\frac{1}{x} + \frac{1}{x^2} \right), \quad x > 0 \quad (39)$$

For the expected value:

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid \theta]] = \mathbb{E}\left[\frac{1}{\theta}\right] = \int_0^1 \frac{1}{\theta} d\theta = \infty \quad (40)$$

so $\mathbb{E}[X]$ is not finite. This aligns with the result from taking the expected value of $f_X(x)$ directly, since an integrand of $\frac{1}{x}$ diverges.

Problem 4: Moment generating function 2

(a) Let $X \sim \text{Exponential}(\lambda)$ with $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.

$$M_X(s) = \mathbb{E}[e^{sX}], \quad (41)$$

$$= \int_0^\infty e^{sx} \lambda e^{-\lambda x} dx, \quad (42)$$

$$= \lambda \int_0^\infty e^{x(s-\lambda)} dx \quad (43)$$

Assuming $s < \lambda$, so that the exponent is negative and the integral converges,

$$M_X(s) = \lambda \left(\frac{1}{s-\lambda} e^{x(s-\lambda)} \right) \Big|_0^\infty = \lambda \left(0 - \frac{1}{s-\lambda} \right) = \frac{-\lambda}{s-\lambda} \quad (44)$$

so the MGF is finite for $s < \lambda$. Differentiating,

$$M'_X(s) = \frac{\lambda}{(s - \lambda)^2}, \quad M'_X(0) = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda} = \mathbb{E}[X], \quad (45)$$

$$M''_X(s) = \frac{-2\lambda}{(s - \lambda)^3}, \quad M''_X(0) = \frac{2}{\lambda^2}, \quad (46)$$

and therefore $\text{Var}(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$.

(b) Let $S_n = X_1 + \dots + X_n$ with X_i i.i.d. $\text{Exponential}(\lambda)$. By independence,

$$M_{S_n}(s) = \left(\frac{-\lambda}{s - \lambda} \right)^n, \quad (47)$$

$$M'_{S_n}(s) = n \left(\frac{-\lambda}{s - \lambda} \right)^{n-1} \cdot \frac{\lambda}{(s - \lambda)^2}, \quad (48)$$

$$M'_{S_n}(0) = n(1)^{n-1} \cdot \frac{1}{\lambda} = \frac{n}{\lambda} = \mathbb{E}[S_n], \quad (49)$$

$$M''_{S_n}(s) = n(n-1) \left(\frac{-\lambda}{s - \lambda} \right)^{n-2} \cdot \frac{\lambda}{(s - \lambda)^2} \cdot \frac{\lambda}{(s - \lambda)^2} + n \left(\frac{-\lambda}{s - \lambda} \right)^{n-1} \cdot \frac{-2\lambda}{(s - \lambda)^3}, \quad (50)$$

$$M''_{S_n}(0) = n(n-1) \cdot \frac{1}{\lambda^2} + n \cdot \frac{2}{\lambda^2} = \frac{n(n-1) + 2n}{\lambda^2} = \frac{n(n+1)}{\lambda^2}, \quad (51)$$

$$\text{Var}(S_n) = \frac{n(n+1)}{\lambda^2} - \frac{n^2}{\lambda^2} = \frac{n}{\lambda^2} \quad (52)$$

This MGF is that of a gamma distribution with shape n and rate λ , so the total waiting time for n exponential events is $\text{Gamma}(n, \lambda)$.

Problem 5: Luck, Skill, and Upsets

(a) With $\sigma(\beta s) = \frac{1}{1 + e^{-\beta s}}$ and $\beta > 0$, the match is a coin flip with probability α and decided by skill otherwise, so

$$f(s) = \alpha \left(\frac{1}{2} \right) + (1 - \alpha) \sigma(\beta s) = \frac{\alpha}{2} + (1 - \alpha) \left(\frac{1}{1 + e^{-\beta s}} \right) \quad (53)$$

Replacing s with $-s$ and multiplying through by $e^{-\beta s}/e^{-\beta s}$,

$$f(-s) = \frac{\alpha}{2} + (1 - \alpha) \left(\frac{1}{1 + e^{\beta s}} \right) \cdot \frac{e^{-\beta s}}{e^{-\beta s}}, \quad (54)$$

$$= \frac{\alpha}{2} + (1 - \alpha) \left(\frac{e^{-\beta s}}{1 + e^{-\beta s}} \right), \quad (55)$$

$$= \frac{\alpha}{2} + (1 - \alpha) \left(\frac{1 + e^{-\beta s}}{1 + e^{-\beta s}} - \frac{1}{1 + e^{-\beta s}} \right), \quad (56)$$

$$= \frac{\alpha}{2} + (1 - \alpha) - (1 - \alpha) \left(\frac{1}{1 + e^{-\beta s}} \right), \quad (57)$$

$$= 1 - \frac{\alpha}{2} - (1 - \alpha) \left(\frac{1}{1 + e^{-\beta s}} \right) = 1 - f(s) \quad (58)$$

For the limits,

$$\lim_{s \rightarrow \infty} f(s) = \frac{\alpha}{2} + (1 - \alpha) = 1 - \frac{\alpha}{2}, \quad \lim_{s \rightarrow -\infty} f(s) = \frac{\alpha}{2} \quad (59)$$

As player i gets infinitely better than player j , the chance he wins approaches 1 minus the chance the game is determined by luck and lost on the coin flip. As player i gets infinitely worse, the only way he wins is on the coin flip, so his chance of winning is the chance the game is determined by luck times one half.

- (b) With $\alpha = 0.2$, $\beta = 1$, and $s = 2$,

$$\mathbb{P}(i \text{ wins}) = f(2) = \frac{0.2}{2} + (1 - 0.2) \left(\frac{1}{1 + e^{-2}} \right) = \frac{1}{10} + \frac{4}{5} \left(\frac{1}{1 + e^{-2}} \right) \approx 0.8046 \quad (60)$$

For a \$1 wager at even odds the bettor receives \$2 on a win and nothing on a loss, so with W the amount returned,

$$\mathbb{E}[W] = 2\mathbb{P}(i \text{ wins}) + 0 \cdot (1 - \mathbb{P}(i \text{ wins})) = 2(0.8046) \approx 1.609 \quad (61)$$

That figure includes the \$1 that was staked, so the expected net return is $1.609 - 1 \approx \$0.6092$ per game. Finally, letting C be the event that the coin decided the match, we have $\mathbb{P}(C) = 0.2$ and $\mathbb{P}(j \text{ wins}) = 1 - 0.8046 \approx 0.1954$, so Bayes' theorem gives

$$\mathbb{P}(C \mid j \text{ wins}) = \frac{\mathbb{P}(j \text{ wins} \mid C) \mathbb{P}(C)}{\mathbb{P}(j \text{ wins})} = \frac{(0.5)(0.2)}{0.1954} \approx 0.5119 \quad (62)$$

- (c) Let E be the event that the weaker player won and F the event that the match was decided by the coin, with $s > 0$. Then

$$\mathbb{P}(E) = 1 - f(s) = \frac{\alpha}{2} + (1 - \alpha) \frac{1}{1 + e^{\beta s}}, \quad (63)$$

$$\mathbb{P}(F) = \alpha, \quad (64)$$

$$\mathbb{P}(E \mid F) = \frac{1}{2} \quad (65)$$

The two events are independent exactly when $\mathbb{P}(E \mid F) = \mathbb{P}(E)$, so α and s must be chosen so that $\mathbb{P}(E) = \frac{1}{2}$:

$$\frac{1}{2} = \frac{\alpha}{2} + (1 - \alpha) \frac{1}{1 + e^{\beta s}}, \quad (66)$$

$$0 = \frac{\alpha - 1}{2} + (1 - \alpha) \frac{1}{1 + e^{\beta s}}, \quad (67)$$

$$0 = (1 + e^{\beta s})(\alpha - 1) + 2(1 - \alpha), \quad (68)$$

$$= \alpha + \alpha e^{\beta s} - 1 - e^{\beta s} + 2 - 2\alpha, \quad (69)$$

$$0 = -\alpha + \alpha e^{\beta s} - e^{\beta s} + 1, \quad (70)$$

$$0 = (\alpha - 1) (e^{\beta s} - 1) \quad (71)$$

The product vanishes when $\alpha = 1$ or when $s = 0$, but since the problem fixes $s > 0$, the second root is unavailable and we only take $\alpha = 1$. So E and F are independent when $(\alpha, s) = (1, s)$ with $s > 0$, and are dependent for every other pair.

(d) Expanding f about $s = 0$ with the Maclaurin series $f(s) = \sum_{i=0}^{\infty} \frac{f^{(i)}(0)}{i!} s^i$:

$$f(0) = \frac{1}{2}, \quad (72)$$

$$f'(s) = -(1-\alpha) \frac{1}{(1+e^{-\beta s})^2} \cdot (-\beta e^{-\beta s}) = (1-\alpha) \frac{\beta e^{-\beta s}}{(1+e^{-\beta s})^2}, \quad (73)$$

$$f'(0) = (1-\alpha) \frac{\beta}{4} \quad (74)$$

and for the second derivative,

$$f''(s) = (1-\alpha) \frac{(1+e^{-\beta s})^2 (-\beta^2 e^{-\beta s}) - \beta e^{-\beta s} \cdot 2(1+e^{-\beta s}) (-\beta e^{-\beta s})}{(1+e^{-\beta s})^4}, \quad (75)$$

$$= (1-\alpha) \frac{-\beta^2 e^{-\beta s} (1+e^{-\beta s}) + 2\beta^2 e^{-2\beta s}}{(1+e^{-\beta s})^3}, \quad (76)$$

$$= (1-\alpha) \frac{\beta^2 e^{-\beta s} (e^{-\beta s} - 1)}{(1+e^{-\beta s})^3}, \quad (77)$$

$$f''(0) = 0 \quad (78)$$

The quadratic term vanishes, so

$$f(s) = \frac{1}{2} + (1-\alpha) \frac{\beta}{4} s + O(s^3) \quad (79)$$

A luck factor α doesn't play a large role across a large number of evenly matched games because as $s \rightarrow 0$, the terms containing α also go to zero, and the only major term remaining is constant $\frac{1}{2}$, which doesn't depend on α .

As s increases or decreases, the luck factor α becomes more influential, eventually being the only parameter that matters as $s \rightarrow \pm\infty$. So, more lopsided matches break this tie.

Problem 6: Simulation Assignment 1: Coupled Inequalities

- (a) Let X, Y, Z be independent Uniform $[0, 1]$. The two conditions are $X^2 + Y^2 < Z$ and $Z^2 > XY$. Since both are lower bounds on z with no upper bound other than $z \leq 1$, so the region is the set of points lying above whichever of the two surfaces is larger.

Working in cylindrical coordinates with $x = r \cos \theta$, $y = r \sin \theta$ and $\theta \in [0, \frac{\pi}{2}]$ (since $x, y \geq 0$), the two bounds become

$$z > r^2, \quad z > \sqrt{r^2 \cos \theta \sin \theta} = r \frac{\sqrt{\sin(2\theta)}}{\sqrt{2}} \quad (80)$$

They exchange dominance where $r^2 = r \sqrt{\cos \theta \sin \theta}$, that is at

$$r^*(\theta) = \sqrt{\cos \theta \sin \theta} = \frac{\sqrt{\sin(2\theta)}}{\sqrt{2}} \quad (81)$$

For $r > r^*$ the parabolic bound r^2 is the larger one, and for $r < r^*$ the $\sqrt{xy}$ bound is. Splitting the radial integral at r^* and including the Jacobian r ,

$$P = \int_0^{\pi/2} \int_{\frac{\sqrt{\sin 2\theta}}{\sqrt{2}}}^1 \int_{r^2}^1 r \, dz \, dr \, d\theta + \int_0^{\pi/2} \int_0^{\frac{\sqrt{\sin 2\theta}}{\sqrt{2}}} \int_{r \frac{\sqrt{\sin 2\theta}}{\sqrt{2}}}^1 r \, dz \, dr \, d\theta, \quad (82)$$

$$= \int_0^{\pi/2} \int_{\frac{\sqrt{\sin 2\theta}}{\sqrt{2}}}^1 (r - r^3) \, dr \, d\theta + \int_0^{\pi/2} \int_0^{\frac{\sqrt{\sin 2\theta}}{\sqrt{2}}} \left(r - r^2 \frac{\sqrt{\sin 2\theta}}{\sqrt{2}} \right) \, dr \, d\theta \quad (83)$$

Carrying out the radial integrals,

$$P = \int_0^{\pi/2} \left[\left(\frac{1}{2} - \frac{1}{4} \right) - \left(\frac{\sin(2\theta)}{4} - \frac{\sin^2(2\theta)}{16} \right) \right] d\theta + \int_0^{\pi/2} \left[\frac{\sin(2\theta)}{4} - \frac{\sin^2(2\theta)}{12} \right] d\theta \quad (84)$$

Using $\sin^2(2\theta) = \frac{1}{2}(1 - \cos(4\theta))$ and clearing denominators,

$$P = \int_0^{\pi/2} \frac{\frac{1-\cos(4\theta)}{2} - 4\sin(2\theta) + 4}{16} d\theta + \int_0^{\pi/2} \frac{3\sin(2\theta) - \frac{1-\cos(4\theta)}{2}}{12} d\theta, \quad (85)$$

$$= \int_0^{\pi/2} \frac{-\cos(4\theta) - 8\sin(2\theta) + 9}{32} d\theta + \int_0^{\pi/2} \frac{6\sin(2\theta) - 1 + \cos(4\theta)}{24} d\theta \quad (86)$$

Evaluating the integrals results in:

$$P = \left(\frac{-\sin(4\theta)}{128} + \frac{\cos(2\theta)}{8} + \frac{9\theta}{32} \right) \Big|_0^{\pi/2} + \left(\frac{-\cos(2\theta)}{8} - \frac{\theta}{24} + \frac{\sin(4\theta)}{96} \right) \Big|_0^{\pi/2}, \quad (87)$$

$$= \left(0 + \frac{-1}{8} + \frac{9\pi}{64} \right) - \frac{1}{8} + \left(\frac{1}{8} - \frac{\pi}{48} \right) + \frac{1}{8}, \quad (88)$$

$$= \frac{9\pi}{64} - \frac{\pi}{48} = \frac{23\pi}{192} \approx 0.3763 \quad (89)$$

So the probability that both inequalities hold is $\frac{23\pi}{192} \approx 0.3763$.